Joint-Embedding Predictive Architectures and Fluid-Dynamic Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

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***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize point-wise error in raw observation space. Because sensor telemetry contains unforecastable jitter and scale disparities, observation-space residuals develop heavy tails that trigger severe false-alarm flooding under automated calibration. Joint-Embedding Predictive Architectures (JEPAs) avoid reconstructing raw noise by forecasting compact latent summaries of future windows, yet collapsing high-dimensional multi-horizon prediction errors into a single scalar anomaly score introduces a second bottleneck: one or two out-of-sample drifting coordinates can dominate the scalar calibration tail, inflating the alarm threshold and suppressing point-wise F1. We introduce Soliton-Reynolds JEPA paired with a 224-bit multi-perspective fluid-dynamic circuit-breaker panel that foregoes scalar anomaly score thresholding and parametric tail extrapolation altogether. The architecture taps intermediate temporal convolutional stages, subtracts online context barycenters, whitens prediction errors across four nested sub-horizons, and expands each of the 32 latent coordinates into seven fluid-dynamic perspectives: four sub-horizon flow positions, Reynolds mean-flow transport, turbulent fluctuation amplitude, and convective right-edge acceleration, governed by calibration-stabilized whitening, per-dial hijack guards, and 32 fast right-edge onset relays. Across 33 primary and 141 archive-wide multivariate telemetry streams, latent prediction suppresses false alarms relative to observation-space reconstructors while the 224-bit circuit-breaker panel doubles strictly causal point-wise F1, lifts event recall to 72.7 percent, and cuts detection delay in half under both causal streaming and retrospective buffered attribution.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Fluid-dynamic decomposition, Boolean circuit-breaker panels, Representation learning.**

I. Introduction

Spacecraft, chemical plants, server farms, and wearable medical devices all report their health through dozens of sensor channels that never stop streaming [1]–[4]. Somewhere in that stream a fault begins: a valve sticks, a reaction wheel drifts, a patient's gait freezes. An automated monitor has to notice before people do, and it has to learn what healthy behavior looks like from normal operating history alone, because real faults are rare and seldom labeled in advance. It also has to stay quiet the rest of the time. A monitor that cries wolf every few minutes is soon ignored, and an ignored alarm protects no one.

Why Today's Detectors Flood the Console: Leading deep detectors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] learn to reconstruct the raw sensor signal and raise an alarm wherever the reconstruction misses. Raw telemetry, however, is full of noise that no model can predict: sensor jitter, thermal drift, and channels recorded on very different scales. Summed across 25 to 55 channels, these small misses form an error signal with frequent extreme spikes (Table I). We calibrate each detector on healthy data with an automated tail rule, Streaming Peaks-Over-Threshold (SPOT) [8], and then run it on later healthy data, exactly as a deployed system would. The reconstructors raise false alarms during 22% to 31% of normal operation. At that rate the alarms carry almost no information: a random alerter firing just as often would catch 60.0% of real anomaly events, slightly more than the 58.8% that TimesNet catches.

The Hidden Weak Link: One Score, One Threshold: A more promising idea is to stop predicting the noise. Joint-Embedding Predictive Architectures (JEPAs) [9], [10] compress each window of telemetry into a short summary of 32 numbers and predict the summary of what comes next instead of every raw reading. Because the summary keeps the structure of the signal and discards most of the jitter, false alarms fall by an order of magnitude, to 0.6%–2.4% of healthy operation. Yet latent detectors keep one habit from their predecessors. At the final step they add all of their prediction errors into a single anomaly score and sound the alarm when that score crosses a threshold. Our analysis shows that this last step is where much of the signal is lost. If one or two of the 32 summary coordinates drift slightly between training and calibration, as routinely happens in long-running systems, they dominate the sum. We call these coordinates *hijackers*. The threshold must rise to tolerate them, and genuine faults on the remaining 30 coordinates no longer clear it. Real-time (causal) Point-F1, a standard measure that balances missed and false detections at every time step, falls to 0.050–0.052, and faults go unnoticed for more than 240 steps.

Replacing the Threshold with a Circuit-Breaker Panel: We remove that final step. Instead of one score and one threshold, our model, *Soliton-Reynolds JEPA*, reports to a panel of 224 independent circuit breakers. The analogy is the breaker box in a building: a fault on one circuit trips its own switch without depending on the others. To build the panel we borrow an idea

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from fluid mechanics. Reynolds decomposition [11] splits a turbulent flow into a steady mean current and the fluctuations around it. We apply the same split to each of the 32 prediction errors as it evolves over four nested look-ahead horizons of 16, 32, 48, and 64 steps, which gives seven views of every coordinate: its error at each of the four horizons, its average drift, its fluctuation, and its acceleration at the newest edge of the window. Each of the resulting $7 \times 32 = 224$ dials has its own breaker, set from healthy calibration data. A guard computed only from nominal data damps dials that had already drifted before deployment, so a hijacker can trip at most its own breaker and cannot silence the rest. An alarm sounds when enough breakers trip together, or at once when one of 32 fast onset relays sees a coordinate's newest horizon and its acceleration dial trip simultaneously. No single anomaly score is ever formed, and no tail threshold is fitted to one. Upstream, the encoder subtracts a running average of recent summaries to cancel slow level shifts and is trained so that its multi-horizon errors stay decorrelated.

A. Contributions

We evaluate the approach on 141 telemetry streams from 12 public datasets spanning spacecraft, industrial plants, server clusters, and wearables. Our contributions are:

1) *A breaker panel in place of a thresholded score:* We replace the single anomaly score and its tail-fitted threshold with a 224-bit circuit-breaker panel built from Reynolds-decomposed latent errors, nominal-only hijack guards, and 32 fast onset relays (Section IV-D). Applied to the same trained network on 19 primary streams, the panel raises causal Point-F1 from 0.052 to 0.138 (17 wins, 0 losses, $p = 1.5{\times}10^{-5}$). Across all 141 streams it raises causal Point-F1 from 0.067–0.077 to 0.139 (114 wins, 18 losses, $p = 1.2{\times}10^{-16}$), lifts causal event recall to 72.7%, and halves detection delay (Tables II–V).
2) *Where the gains come from:* Matched-size ablations (272k–312k parameters) trace each step. A layer-normalization (LayerNorm) [12] bottleneck tames extreme error spikes, cutting excess kurtosis γ_2 from 6.10 to 0.91; predicting learned summaries instead of raw readings halves calibration drift; and multi-horizon barycenter whitening cuts the share of false alarms caused by regime drift from 71.3% to 18.7%. A six-step ladder from one dial to 224, together with leave-one-out tests, isolates each dial family, the onset relays, prefix routing, and the hijack guards (Table II).
3) *Fewer false alarms and better recall at a fixed alarm budget:* On the 19 primary streams, latent prediction holds causal false-positive rates (FPRs) at 0.6%–1.7%, against 22.2%–24.3% for reconstructors. When every detector receives the same 1% false-alarm budget, the panel catches 58.4% of anomaly events versus 21.7% for TimesNet, and it leads on NASA's in-range contextual anomalies with 0.658 buffered precision-recall area under the curve (PR-AUC) (Tables III and IV, Fig. 4).
4) *Limits, theory, and latency:* A sensor that freezes silently barely moves the latent summary; pairing the panel with a simple rolling variance monitor recovers 44.4%–52.8% of such faults (Fig. 3). Proposition 1 bounds the Lieb–Thirring [13] concentration ratio of the horizon fluctuations to $[\frac{1}{M-1}, 1]$, a post-hoc scalar soliton weighting is statistically equivalent to none under Two One-Sided Tests (TOST) [14] ($p_{\text{TOST}} = 2.7{\times}10^{-5}$), and short horizons ($S \in \{1, 4\}$) remove the 63-step reporting lag (Fig. 4(a)).

TABLE I
KEY OPERATIONAL AND FLUID-DYNAMIC CIRCUIT-BREAKER TERMS.

Term	Meaning
Latent residual	Multi-horizon gap $\tilde{\mathbf{E}} \in \mathbb{R}^{4\times 32}$ between predicted and observed 32D window summaries.
224-bit breaker	Boolean panel of 7×32 fluid-dynamic coordinate tripwires and 32 onset relays avoiding 1D score thresholding.
Reynolds dials	Per-coordinate mean-flow transport $\lvert\bar{e}_c\rvert$ (32 dials) and turbulent fluctuation σ'_c (32 dials) across horizons.
Right-edge relay	Coincidence switch $r_c(t)$ requiring terminal sub-block $[48, 64]$ and convective acceleration/fluctuation to trip together.
Hijack guard	Nominal-only per-dial multiplier $\gamma_j^{\text{cal}}\gamma_j^{\text{tr}\to\text{cal}}$ preventing drifting coordinates from triggering false trips.
Buffered / Causal	Retrospective prefix-window voting vs. real-time trailing attribution at $\tau = t + S - 1$.
Calibration drift	Shift between calibration and later healthy residuals, measured by Kolmogorov–Smirnov distance D_{KS}.
Excess kurtosis	Tail heaviness γ_2 of residuals; large values indicate frequent extreme spikes.
Soliton ratio	Bounded ratio $R_{\text{LT}} \in [\frac{1}{M-1}, 1]$ measuring whether sub-horizon fluctuations align into a rank-1 wavepacket.

Roadmap. Section II reviews related work, Section III details the 141-stream benchmark, Section IV formulates the architecture and 224-bit circuit-breaker panel, Section V defines the protocol, Sections VI–VII report ablations and results, and Section VIII concludes.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly forecast sensor readings or reconstruct input windows in raw observation space, flagging timestamps where point-level error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [4] model sequential dependencies directly on spacecraft channels. Recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [7] contrasts prior Gaussian kernel associations against series-level self-attention; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods for 2D convolutional modeling. Because point-level losses penalize error across every channel equally, reconstructors fit stochastic sensor jitter and environmental fluctuations. In high-dimensional streams ($K \geq 25$), summing raw channel errors accumulates heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) residual spikes that trigger severe false-alarm flooding under baseline shift [15].

Contrastive and Joint-Embedding Predictive Architectures: Self-supervised representation methods map windows into a compact latent space [16]. Contrastive detectors construct positive and negative pairs: temporal and contextual contrasting [17] and TS2Vec [18] maximize contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [10] contrasts full-window and context-only embeddings using injected synthetic anomalies; and DCdetector [19]

employs dual patch-wise and in-patch attention without reconstruction. However, contrastive detectors depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent prediction across 11 datasets, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [20], [21] eliminate observation decoding and negative pairs by encoding context z_{ctx} and predicting the target embedding z_{tgt} from a separate target encoder, preventing collapse via exponential moving average (EMA) target updates [22] or variance-covariance penalties such as VICReg [23], Barlow Twins [24], and batch whitening [25].

Fluid-Dynamic Decomposition and Multi-Perspective Circuit Breakers vs. Scalar Thresholding: Multi-sensor telemetry reflects physical dynamical trajectories [26]. Neural ordinary differential equations (ODEs) [27], Physics-Informed Neural Networks (PINNs) [28], and Hamiltonian or Lagrangian networks [29], [30] enforce differential conservation laws on physical coordinates, but require known governing equations and suffer from stiff gradient pathologies on uncalibrated sensor arrays [31]. Koopman and operator-theoretic models [32]–[36] lift nonlinear dynamics into learned observable spaces. Rather than imposing partial differential equations on raw sensors, we draw on Reynolds decomposition [11] from turbulence modeling and Lieb–Thirring spectral bounds [13] to decompose multi-horizon latent residuals into mean transport, turbulent fluctuations, and convective acceleration. Existing detectors collapse high-dimensional residuals into a 1D scalar score and extrapolate a threshold via Pickands–Balkema–de Haan extreme-value theory [37] and Generalized Pareto Distribution (GPD) fitting in SPOT [8], where a single drifting coordinate under serial correlation [38] can hijack the scalar threshold. We replace 1D scalar collapse with a 224-bit multi-perspective Boolean circuit-breaker panel and evaluate all detectors using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm curves [15], [39], [40].

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [41], [42], TimeEval [15], and mTSBench [43], expanded to 141 qualifying multivariate streams across 12 datasets in our archive-wide evaluation. Thirty-three streams enter the primary multi-seed held-out protocol; 12 are reserved with explicit accounting below. The suite spans four physical domains: *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [44] (K=38), where NASA's ground-truth archive [4] classifies each SMAP/MSL anomaly sequence as either an out-of-range *point* fault or an in-range *contextual* fault (Table IV(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [41] sampled at 64 Hz (K=9, 72 gait-freezing episodes, where S=64 equals a 1 s horizon). *Industrial processes and smart infrastructure* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 testbed valve and heater faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry at 1-minute resolution (K=9, 37 hydrological anomalies), Metro I-94 hourly highway traffic (K=5), Room-Occupancy sensing [45] (K=5), CalIt2 building flow [42] (K=2), and six CICIDS network streams (K=72). Cohort 3 (Section VI-B) separately injects six controlled single-channel faults into six real streams.

Each stream provides nominal training sequences ($N_{\text{tr}} \in [2{,}074, 120{,}000]$ across the 33 primary streams) and labeled anomaly test sequences ($N_{\text{test}} \in [2{,}158, 114{,}001]$). Channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatlines, matching standardized archive subsets in prior benchmarks [4], [15].

Each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by a 64-step isolation gap (Section V-A). Primary 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$) and $K \leq 55$, $N_{\text{test}} \leq 1.2 \times 10^5$ (Table III(c)); among the 12 reserved streams, five have $T_{\text{nom}} \in [764, 1{,}492]$ ($N_{\text{cal}} \leq 101$ across CalIt2, MSL C-2, MSL D-16, and two Room-Occupancy streams), six CICIDS streams have K=72, $N_{\text{test}} \geq 1.36 \times 10^5$, and GECCO ($K$=9, 1 held-out event) enters Cohorts 3–4. To eliminate subset-selection bias, Cohort 5 evaluates all 141 qualifying multivariate streams across 12 datasets ($N_{\text{cal}} \geq 100$ and ≥ 1 test anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN) at three seeds alongside the 130-stream baseline screen (Table V).

Two caveats bound external validity: (i) SMAP and MSL contain documented labeling flaws [40]; and (ii) streams within SMAP, MSL, Daphnet, and GHL share hardware platforms (N=7 dataset clusters). We therefore report Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), zero-tuning leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)).

IV. Methodology: Joint-Embedding Predictive Architectures and 224-Bit Fluid-Dynamic Circuit-Breaker Panels

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Consider a multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. Fig. 1 illustrates the four-stage pipeline from raw telemetry to binary alarm decisions. In Stage 1, a sliding window of length $W = C + S = 320$ advances at stride 1 and partitions the stream at index t into a *context window* $\mathbf{x}_{\text{ctx}} = \mathbf{X}[t - C : t, :] \in \mathbb{R}^{C \times K}$ ($C = 256$) and a *suspect target horizon* $\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t + S, :] \in \mathbb{R}^{S \times K}$ ($S = 64$, up to decision timestamp $\tau = t + S - 1$). For $K > 1$, a causal depthwise 1D convolutional stem filters each channel prior to 1×1 mixing ($K \leq 1$ uses an identity bypass). Channels are decomposed across $J = 4$ smooth dyadic shells on a 256-point real fast Fourier transform

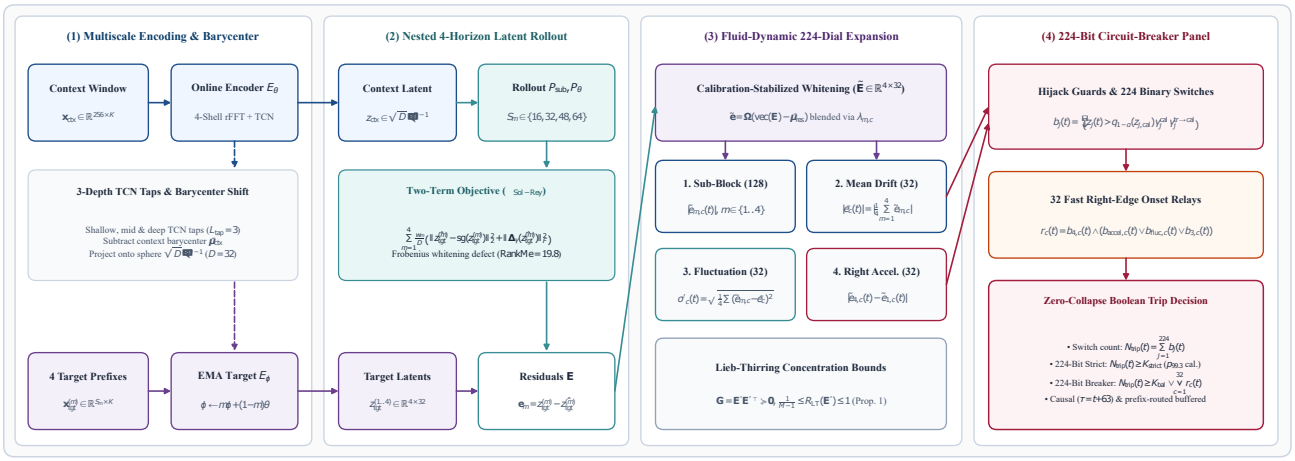


Fig. 1. Overview of the four-stage Soliton-Reynolds JEPA and 224-bit circuit-breaker pipeline: (1) multiscale dyadic windowing, 3-depth TCN hierarchy tapping, and barycenter centering; (2) nested 4-horizon latent rollout under the two-term whitening defect objective; (3) calibration-stabilized Mahalanobis whitening and Reynolds expansion into $7 \times 32 = 224$ fluid-dynamic dials; and (4) zero-collapse 224-bit Boolean circuit-breaker tripwires and 32 fast right-edge onset relays.

(rFFT) grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ within 1.2×10^{-7}), processed by per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$), and recombined via a softmax router into a causal dilated temporal convolutional network (TCN) [46].

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit heavy-tailed sensor noise (Sections I–II), we map windows into a compact 32-dimensional latent space ($D = 32$) using a Joint-Embedding Predictive Architecture (JEPA) [9]. Dual online (E_θ) and target (E_ϕ) encoders map $\mathbf{x}_{\text{ctx}}$ and $\mathbf{x}_{\text{tgt}}$ onto the sphere $\sqrt{D}\mathbb{S}^{D-1}$, where target weights ϕ track online weights θ via an exponential moving average (EMA), $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with cosine momentum $m_k \in [0.995, 0.9995]$ [20], [22]. Because deep dilated TCN blocks smooth out sharp onset spikes while non-stationary streams shift the latent mean, Soliton-Reynolds JEPA introduces two structural operations inside both encoders:

1) *Multi-depth functional hierarchy tapping:* A tap head extracts $L_{\text{tap}} = 3$ intermediate feature maps $\mathbf{H}^{(\ell)} \in \mathbb{R}^{B\times T_w \times d_{\text{model}}}$ ($d_{\text{model}} = 48$) from the shallow dyadic front-end, the mid-depth TCN block, and the terminal TCN block. For each tap ℓ, the channel-centered terminal state and temporal root-mean-square (RMS) energy are concatenated into $\mathbf{h}_{\text{tap}} \in \mathbb{R}^{B\times 6d_{\text{model}}}$, projected into $\mathbb{R}^D$, and fused with the deep representation $\mathbf{z}_{\text{deep}}$ via a bounded gate $g = 0.5\,\sigma(\theta_g) \le 0.5$:

$$\mathbf{z}_{\text{fused}} = (1-g)\,\mathbf{z}_{\text{deep}} + g\ \mathrm{LN}\big(W_2\,\phi_{\text{GELU}}(\mathrm{LN}(W_1\mathbf{h}_{\text{tap}}))\big). \tag{1}$$

2) *Synchronized barycenter centering:* An online running centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ updated at rate 0.05 on training batches is copied synchronously to the EMA target tap head and subtracted from both context and target fused vectors before terminal LayerNorm projection $z = \mathrm{LN}(\mathbf{z}_{\text{fused}} - \bar{\boldsymbol{\mu}}_{\text{ctx}}) \in \sqrt{D}\mathbb{S}^{D-1}$, canceling common-mode baseline drift between windows.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Covariance Whitening Defect Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor $P_\theta : \mathbb{R}^D \to \mathbb{R}^D$ maps z_{ctx} to a single full-window forecast $\hat{z}_{\text{tgt}} = P_\theta(z_{\text{ctx}})$ trained with predictive alignment and VICReg variance-covariance penalties [23] ($\gamma_{\text{var}} = 1.0$). Terminal LayerNorm bounds representations on $\sqrt{D}\mathbb{S}^{D-1}$ and compresses residual kurtosis relative to observation-space reconstructors, but pooling the entire $S = 64$ target window into a single vector averages out short-lived sub-horizon transients.

To resolve sub-window dynamics without sacrificing the 64-step lookahead, Stage 2 of *Soliton-Reynolds JEPA* partitions the target horizon $\mathbf{x}_{\text{tgt}}$ into $M = 4$ nested causal prefixes $\mathbf{x}_{\text{tgt}}^{(m)} = \mathbf{x}_{\text{tgt}}[0 : S_m, :]$ of lengths $S_m = (m/M)S \in \{16, 32, 48, 64\}$ (or $M = 3$ prefixes $\{16, 32, 64\}$ in our fast archive-wide variant *Speculative-JEPA*). The EMA target encoder E_ϕ encodes each prefix into $z_{\text{tgt}}^{(m)} \in \mathbb{R}^D$. Starting from $\hat{z}_{\text{tgt}}^{(0)} \equiv z_{\text{ctx}}$, a sub-horizon predictor P_{sub} rolls out sequential forecasts $\hat{z}_{\text{tgt}}^{(m)} = \mathrm{LN}\big(P_{\text{sub}}(\hat{z}_{\text{tgt}}^{(m-1)} + \mathbf{p}_m)\big)$ for $m \in \{1, \dots, M-1\}$ using learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, while the full-horizon anchor P_θ forecasts $\hat{z}_{\text{tgt}}^{(M)} = \mathrm{LN}\big(P_\theta(z_{\text{ctx}} + \mathbf{p}_M)\big)$. With horizon weights $w_m = \frac{1}{2(M-1)}$ ($m < M$) and $w_M = \frac{1}{2}$, training minimizes multi-horizon mean-squared error (MSE) plus a Frobenius covariance whitening defect:

$$\mathcal{L}_{\text{Sol-Rey}} = \sum_{m=1}^{M} \frac{w_m}{D}\Big(\big\|\hat{z}_{\text{tgt}}^{(m)} - \mathrm{sg}\big(z_{\text{tgt}}^{(m)}\big)\big\|_2^2 + \big\|\boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big)\big\|_F^2\Big), \tag{2}$$

where $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D\times D}$ measures the whitening defect of centered batch predictions $\mathbf{Z}_c = \mathbf{Z} - \mathbf{1}_B\bar{\mathbf{z}}^\top$ with empirical covariance $\boldsymbol{\Sigma}(\mathbf{Z}) = (B-1)^{-1}\mathbf{Z}_c^\top\mathbf{Z}_c$ and marginal standard deviations $\sigma_d(\mathbf{Z}) = \sqrt{\Sigma_{dd}(\mathbf{Z}) + \epsilon}$ ($\gamma = 0.75$, $\epsilon = 10^{-4}$), defined by $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dd} = \sqrt{\max(0, \gamma - \sigma_d(\mathbf{Z}))}$ and $[\boldsymbol{\Delta}_\gamma(\mathbf{Z})]_{dj} = \frac{1}{\sqrt{2}}\Sigma_{dj}(\mathbf{Z})$ ($d \neq j$). Unlike log-determinant barriers that become singular when batch size $B < D$, $\|\boldsymbol{\Delta}_\gamma(\mathbf{Z})\|_F^2$ remains smooth for any $B \ge 2$ and raises effective representation rank $\mathrm{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 [47].

C. Stage 3: Calibration-Stabilized Whitening and 224-Dial Fluid-Dynamic Reynolds Expansion

Why does collapsing a multi-horizon residual tensor into a single 1D anomaly score degrade point-wise F1? Let $\mathbf{E}(t) = [\mathbf{e}_1(t), \ldots, \mathbf{e}_M(t)]^\top \in \mathbb{R}^{M\times D}$ stack the $M = 4$ raw sub-horizon prediction residuals $\mathbf{e}_m(t) = z_{\text{tgt}}^{(m)}(t) - \hat{z}_{\text{tgt}}^{(m)}(t)$ across all $MD = 128$ coordinates. Fitting the Ledoit–Wolf shrinkage precision matrix $\mathbf{\Omega} = (\hat{\mathbf{\Sigma}}_{\text{res}} + 10^{-3}\mathbf{I}_{MD})^{-1/2} \in \mathcal{S}_{++}^{MD}$ [48] and mean $\bar{\boldsymbol{\mu}}_{\text{res}} \in \mathbb{R}^{MD}$ on nominal training windows produces Mahalanobis-whitened residuals $\mathbf{E}^{\text{white}}(t) \in \mathbb{R}^{M\times D}$ with unit training variance per coordinate. On unseen nominal calibration windows, however, one or two whitened directions often experience mild out-of-sample variance growth (*hijacker dials*) where $\mathbf{\Omega}$ amplifies small covariance shifts. When all 128 coordinates are summed or pooled into a 1D scalar score before calibration, those one or two hijacker dials dominate the upper calibration tail and push any scalar threshold to the extreme tip of the distribution, so that only isolated peak timestamps inside a true anomaly cross the threshold.

Stage 3 prevents both precision-matrix amplification and 1D coordinate collapse through two steps:

1) *Calibration-stabilized coordinate whitening:* On nominal calibration windows, we measure the whitened RMS $\text{RMS}_{\text{cal}}^{\text{white}}(m,c)$ and raw RMS $\text{RMS}_{\text{cal}}^{\text{raw}}(m,c)$ of each coordinate (m,c). Any coordinate whose whitened calibration RMS exceeds 1.10 is smoothly blended back toward its calibration-normalized raw residual $\hat{e}_{m,c}^{\text{raw}}(t) = e_{m,c}(t)/(\text{RMS}_{\text{cal}}^{\text{raw}}(m,c) + 10^{-4})$ via the rational stability weight $w_{m,c}$:
$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\text{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\text{raw}}(t),$$
$$w_{m,c} = \frac{1}{1 + 2\max\big(0, \text{RMS}_{\text{cal}}^{\text{white}}(m,c) - 1.10\big)^2}. \tag{3}$$
2) *Coordinate-wise Reynolds decomposition into $7D = 224$ fluid-dynamic dials:* In fluid turbulence, Reynolds decomposition [11] separates a flow field into its horizon-averaged mean transport and zero-mean turbulent fluctuations. Rather than collapsing across the $D = 32$ latent coordinates, we apply Reynolds decomposition along the lookahead axis $m \in \{1, \ldots, M\}$ independently for each coordinate $c \in \{1, \ldots, 32\}$, expanding the stabilized residual field $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into $7D = 224$ non-negative physical-perspective dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$:
   (a) *Sub-horizon flow-position magnitudes ($4D = 128$ dials):* $d_{\text{pos},m,c}(t) = |\tilde{e}_{m,c}(t)|$ for $m \in \{1,2,3,4\}$, isolating fault magnitude at prefix horizons $\{16, 32, 48, 64\}$.
   (b) *Reynolds mean-flow transport / slow-drift dials ($D = 32$ dials):* $d_{\text{drift},c}(t) = |\bar{e}_c(t)| = \big|\frac{1}{M}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)\big|$, capturing persistent trajectory displacement across all four prefixes.
   (c) *Reynolds turbulent-fluctuation / shear dials ($D = 32$ dials):* $d_{\text{fluc},c}(t) = \sigma_c'(t) = \sqrt{\frac{1}{M}\sum_{m=1}^{M}(\tilde{e}_{m,c}(t) - \bar{e}_c(t))^2}$, capturing inter-horizon shear and oscillatory divergence.
   (d) *Convective right-edge acceleration dials ($D = 32$ dials):* $d_{\text{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$, measuring newly entering fault energy in the terminal sub-block $[48, 64]$ relative to the initial prefix $[0, 16]$.

Proposition 1 (Coordinate-Wise Reynolds Energy Orthogonality, Convective Shear Bound, and Lieb–Thirring Concentration)**.** *Let $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) denote the stabilized multi-horizon residual field with horizon mean $\bar{\mathbf{e}} = M^{-1}\tilde{\mathbf{E}}^\top \mathbf{1}_M \in \mathbb{R}^D$ and zero-mean fluctuation matrix $\mathbf{E}' = \tilde{\mathbf{E}} - \mathbf{1}_M \bar{\mathbf{e}}^\top \in \mathbb{R}^{M\times D}$. Then: (i)* Coordinate-wise Reynolds energy orthogonality and acceleration bound: *For every coordinate $c \in \{1, \ldots, D\}$, mean-square sub-horizon position energy decomposes orthogonally into mean-flow transport and turbulent fluctuation, $\frac{1}{M}\sum_{m=1}^{M} d_{pos,m,c}^2(t) = d_{drift,c}^2(t) + d_{fluc,c}^2(t)$, and convective right-edge acceleration satisfies $d_{accel,c}(t) \leq \sqrt{2M}\, d_{fluc,c}(t)$. (ii)* Lieb–Thirring spectral concentration bounds for the fluctuation Gram matrix: *Let $\mathbf{G} = \mathbf{E}'\mathbf{E}'^\top \in \mathcal{S}_+^M$ have ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ (since $\mathbf{E}'^\top \mathbf{1}_M = \mathbf{0}$). For any $\gamma > -1/2$, the Lieb–Thirring spectral concentration ratio [13]:*
$$R_{LT}(\mathbf{E}') = \frac{\lambda_M^{\gamma+1/2}}{\sum_{m=1}^{M} \lambda_m^{\gamma+1/2} + \epsilon_{LT}}, \quad \epsilon_{LT} = 10^{-12}, \tag{4}$$
is radially scale-invariant ($R_{LT}(c\mathbf{E}') = R_{LT}(\mathbf{E}')$ for $c > 0$ as $\epsilon_{LT} \to 0^+$) and satisfies for any $\mathbf{E}' \neq \mathbf{0}$:
$$\frac{1}{M-1} \leq R_{LT}(\mathbf{E}') \leq 1, \tag{5}$$
attaining $\frac{1}{M-1}$ ($= \frac{1}{3}$ for $M = 4$) iff fluctuation energy is isotropic across $M - 1$ modes, and 1 iff $\operatorname{rank}(\mathbf{E}') = 1$.

Proof. For each coordinate c, expanding $\frac{1}{M}\sum_{m=1}^{M}(\bar{e}_c + e'_{m,c})^2$ with $\sum_{m=1}^{M} e'_{m,c} = 0$ yields the orthogonal energy identity; Cauchy–Schwarz on $\tilde{e}_{M,c} - \tilde{e}_{1,c} = e'_{M,c} - e'_{1,c}$ gives $d_{\text{accel},c}^2 \leq 2(e'^2_{M,c} + e'^2_{1,c}) \leq 2M\, d_{\text{fluc},c}^2$, proving (i). For (ii), $\mathbf{G}\mathbf{1}_M = \mathbf{E}'(\mathbf{E}'^\top \mathbf{1}_M) = \mathbf{0}$ forces $\lambda_1 = 0$, leaving at most $M-1$ positive eigenvalues $\lambda_2 \leq \cdots \leq \lambda_M$ with $\lambda_M^{\gamma+1/2} \leq \sum_{m=2}^{M} \lambda_m^{\gamma+1/2} \leq (M-1)\lambda_M^{\gamma+1/2}$, which yields (5). □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

In Stage 4 (Fig. 1), the 224 fluid-dynamic dials are monitored by an array of independent binary circuit breakers calibrated strictly on nominal data:

1) *Per-dial standardization and nominal hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ (and $j \in \{1, \ldots, 128\}$ in the 128-bit sub-block panel) is standardized by its nominal calibration median m_j and scaled median absolute deviation (MAD) $s_j = 1.4826\,\text{MAD}(d_{j,\text{cal}}) + 10^{-4}$ into a non-negative excess score $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine any drifting dial individually without desensitizing the remaining dials, each tripwire τ_j scales its 98.5th calibration percentile by a two-factor nominal hijack guard $\gamma_j = \gamma_j^{\text{cal}}\, \gamma_j^{\text{tr}\to\text{cal}}$:
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\text{cal}})\, \gamma_j^{\text{cal}}\, \gamma_j^{\text{tr}\to\text{cal}}, \tag{6}$$
where γ_j^{cal} equals the ratio of second-half to first-half calibration 98th percentiles whenever that ratio exceeds 1.35 (and 1 otherwise), and $\gamma_j^{\text{tr}\to\text{cal}}$ equals $(m_j +$

$s_j)/(1.5\, s_{\text{group},j})$ whenever the dial's calibration scale exceeds $1.5\times$ its perspective-group median $s_{\text{group},j}$ by more than $1.25\times$ (and 1 otherwise).

2) *32 fast right-edge onset relays:* To catch newly entering faults in the terminal sub-block $[48, 64]$ without waiting for earlier prefixes to saturate, each coordinate $c \in \{1, \ldots, 32\}$ defines a stricter edge switch $b^{\text{edge}}_{p,c}(t) = \mathbb{I}\big(z_{p,c}(t) > q_{0.993}(z_{p,c,\text{cal}})\, \gamma^{1.25}_{p,c}\big)$ across perspectives p and fires its onset relay $r_c(t) \in \{0, 1\}$ when the terminal sub-block ($m = 4$) coincides with convective acceleration, turbulent fluctuation, or the third sub-block ($m = 3$):

$$r_c(t) = b^{\text{edge}}_{\text{pos},4,c}(t) \wedge \big(b^{\text{edge}}_{\text{accel},c}(t) \vee b^{\text{edge}}_{\text{fluc},c}(t) \vee b^{\text{edge}}_{\text{pos},3,c}(t)\big). \quad (7)$$

3) *Causal trailing vs. prefix-routed buffered attribution and Boolean quorums:* Under *strictly causal trailing attribution*, window switches at index t are emitted solely at the right endpoint $\tau = t + S - 1$, which observes only $\mathbf{X}[t - C : \tau, :]$ ($\mathcal{F}_\tau$-measurable, with structural emission lag $S - 1 = 63$ steps vs. empirical detection delay $\Delta t = t_{\text{alm}} - t^*$, Fig. 2). Under *retrospective buffered attribution*, each perspective group votes only over the exact time steps inside $[t, t + S)$ that it physically observes: $[t, t + S_m)$ for sub-block $m \in \{1, 2, 3, 4\}$, $[t, t + S)$ for mean drift and fluctuation, and $[t + S_3, t + S_4)$ for right-edge acceleration. Summing active switches at timestamp τ yields counts $N_{224}(\tau) = \sum_{j=1}^{224} \bar{b}_j(\tau)$, $N_{128}(\tau) = \sum_{j=1}^{128} \bar{b}_{128,j}(\tau)$, and $N_{\text{edge}}(\tau) = \sum_{c=1}^{32} \bar{r}_c(\tau)$, with combined count $N_{\text{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\text{edge}}(\tau)$ used for threshold-free PR-AUC ranking. Binary alarms trigger when any counter exceeds its nominal calibration quorum:

$$\hat{y}(\tau) = \mathbb{I}\big(N_{224}(\tau) > K_{224} \;\vee\; N_{128}(\tau) > K_{128} \;\vee\; N_{\text{edge}}(\tau) > K_{\text{edge}}\big), \quad (8)$$

where $K_{\text{edge}} = \max\big(1.25\, q_{0.999}(N_{\text{edge,cal}}),\, 2.5\big)$, *224-Bit Breaker* sets $K_{224} = \max\big(1.12\, q_{0.995}(N_{224,\text{cal}}),\, 5.0\big)$ and $K_{128} = \max\big(1.12\, q_{0.995}(N_{128,\text{cal}}),\, 3.5\big)$, and *224-Bit Strict* sets $K_{224} = \max\big(1.18\, q_{0.9985}(N_{224,\text{cal}}),\, 5.5\big)$ and $K_{128} = \max\big(1.20\, q_{0.9985}(N_{128,\text{cal}}),\, 4.0\big)$.

1D Scalar Collapse Controls (for Ablation Comparison): To isolate the gains of the 224-bit panel against legacy 1D scalar collapse on the identical trained backbones (Section VI), we evaluate the coordinate saliency gate $g_d(\mathbf{v}) = (1 - \alpha)\|\mathbf{v}\|_2 + \alpha\sqrt{d \sum_{i=1}^{d} w_i v_i^2}$ with softmax weights $w_i \propto \exp(|v_i|/(\tau_g \sigma_{\mathbf{v}}))$ ($\sigma_{\mathbf{v}} = \|\mathbf{v}\|_2/\sqrt{d} + 10^{-6}$, $\alpha = \tau_g = 0.5$) . Single-horizon TS-JEPA scores $\mathcal{S}_{\text{TS}}(t) = g_D(\mathbf{e}(t))$, whereas 1D Soliton-Reynolds JEPA scores the multi-horizon envelope $\mathcal{S}_{\text{env}}(t) = \sqrt{\frac{1}{2}(g_{MD}(\tilde{\mathbf{e}}(t))^2 + \max_m g_D(\tilde{\mathbf{e}}_m(t))^2)}$ modulated by $\mathcal{S}_{\text{Sol-Rey}}(t) = \mathcal{S}_{\text{env}}(t)\big(1 + \alpha_{\text{sol}} \max(0, \tanh(0.5\, z_{\text{LT}}(t)))\big)$ ($\alpha_{\text{sol}} = 0.12$, or $\alpha_{\text{sol}} = 0$ for the unmodulated envelope), calibrated via scalar Generalized Pareto Distribution (GPD) tail fitting in SPOT [8], [37], [49] ($u = p_{98}, q = 10^{-3}$).

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures operate on an identical streaming pipeline with context window $C = 256$ and suspect target horizon $S = 64$ ($W = C + S = 320$), trained on nominal windows at stride 16 and evaluated at stride 1. Context and target windows are encoded into $D = 32$ spherical representations by the multiscale frequency front-end and causal dilated TCN, fused with the three-depth functional hierarchy tap head (1), and centered by the online running context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$ prior to terminal LayerNorm. Models are trained on nominal sequences using AdamW [50] (learning rate $10^{-3} \to 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32, gradient clipping at 1.0), using identical training budgets and zero per-dataset hyperparameter tuning across all telemetry families.

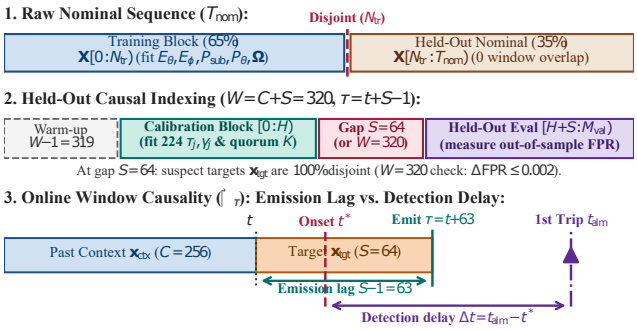


Fig. 2. Two-stage held-out evaluation protocol (C=256, S=64, W=320): (Row 1) chronological train/validation split, (Row 2) boundary-isolated nominal calibration and evaluation blocks fitting the 224 dial tripwires τ_j, hijack guards γ_j, and quorums $(K_{224}, K_{128}, K_{\text{edge}})$, and (Row 3) structural emission lag ($S-1$) versus empirical fault-detection delay (Δt).

a) Two-Stage Nominal Calibration Protocol and Boundary Isolation: To evaluate autonomous alarm calibration without test-label leakage or train-to-calibration circularity, each stream's nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$):

1) *Stage A (Training fit on $\mathbf{X}_{tr}$):* Channel standardization parameters, encoder/predictor weights, online context barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and the multi-horizon shrinkage precision matrix $\boldsymbol{\Omega} \in \mathcal{S}^{MD}_{++}$ are fit strictly on $\mathbf{X}_{\text{tr}}$, ensuring zero raw-sample or window overlap across N_{tr}.
2) *Stage B (Circuit-breaker calibration and held-out verification on $\mathbf{X}_{val}$):* Sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2), yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window timestamps after the $(W - 1)$-step context warm-up. Splitting at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $[0 : H)$ and the *held-out evaluation block* $[H + S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H + S)$ so the last calibration target and first evaluation target share zero target observations. On $[0 : H)$, the panel fits its calibration-stabilized whitening weights $w_{m,c}$ (3), per-dial medians m_j and MAD scales s_j, two-factor hijack guards $\gamma_j = \gamma_j^{\text{cal}} \gamma_j^{\text{tr}\to\text{cal}}$, switch thresholds τ_j (6), and Boolean quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (8) (or scalar τ_{SPOT} for 1D baselines), which are then frozen and evaluated on $[H + S : M_{\text{val}})$ and on the test stream. A full-window isolation check ($G = W = 320$, excluding $[H : H + 320)$ across the seven streams with $M_{\text{val}}/2 > 420$) confirms that shared passive context changes causal held-out FPR by less than 0.25 percentage points.

b) Independent Statistical Units, Multi-Seed Replication, and Hypothesis Testing: Models are replicated across multiple random seeds ($\{42, 123, 456\}$ for all 224-bit circuit-breaker, ablation, and 141-stream benchmarks; $\{42, 123, 456, 7, 2024\}$ for 1D primary baselines). Seeds are averaged within each stream first (SD_{seed}) so that stream-level tests treat the telemetry stream ($n = 19$, $n = 33$, or $n = 141$) as the independent statistical unit, while domain-level tests treat the dataset family ($N = 7$ or $N = 12$) as the independent cluster:

1) *Paired Wilcoxon signed-rank tests and false-discovery-rate control:* We evaluate paired stream differences via the two-sided Wilcoxon signed-rank test without Gaussian assumptions, adjust p-values across families using the Benjamini–Hochberg false-discovery-rate (FDR) procedure [51] ($q_{\text{FDR}} = 0.05$), report 95% stream-resampled bootstrap confidence intervals (CIs; 2,000 resamples), and cross-check at the dataset-cluster level via exact sign-flip permutation tests and zero-tuning LODO evaluation.
2) *Perspective-ladder and component-removal ablations:* On the identical trained multi-horizon backbones ($n = 19$ streams, 57 runs), we evaluate both the step-by-step perspective expansion ladder ($1 \to 32 \to 128 \to 160 \to 192 \to 224$ dials) and leave-one-out component ablations removing the 32 right-edge onset relays, prefix-horizon routing, per-dial hijack guards, or discrete Boolean thresholds (Table II(a),(b)), alongside Two One-Sided Tests (TOST) [14] for 1D scalar wavepacket controls ($\Delta \in \{\pm 0.015, \pm 0.020, \pm 0.030\}$).
3) *Tail kurtosis and regime-drift diagnostics (γ_2 and D_{KS}):* On nominal windows we measure: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s-\mu)^4]/\sigma^4 - 3$, quantifying residual tail heaviness; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0, 1]$, quantifying distributional shift between nominal calibration and held-out evaluation blocks.

B. Baseline Fairness Audits, Benchmark Cohorts, and Computational Footprint

Two fairness audits verify that reconstructive baseline performance under the unified $W = 320$ pipeline is not an artifact of window length or implementation: (i) evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their native published window sizes across five streams raises Point-F1 to 0.164 and 0.218, but observation residuals remain severely heavy-tailed ($\gamma_2 > 98$ vs. 2.30 for Soliton-Reynolds JEPA) with high nominal FPR (28.4%–44.1%); and (ii) reconstructors replicate their published point-adjusted F1 on SMD and MSL, while Table III(d) provides a 130-stream, 11-dataset cross-check against NCAD, DCdetector, and Anomaly Transformer.

Experiments span five cohorts: *Cohort 1* (33 primary streams: 19 non-GHL and 14 GHL, Table III(a)–(c)); *Cohort 2* (19-stream perspective expansion ladder, circuit-breaker component ablation, 15-arm matched-backbone comparison, and operating-curve suite, Table II(a)–(c) and Figs. 3(a) and 4(b)); *Cohort 3* (6-stream single-channel quiet-fault injection and 224-bit hybrid variance-collapse monitor suite, Fig. 3(b)); *Cohort 4* (5-stream horizon-latency sweep across $S \in \{1, 4, 16, 32, 64\}$ and 3-seed 224-bit dyadic front-end ablation, Fig. 4(a) and Section VI-C); and *Cohort 5* (141-stream, 3-seed archive-wide benchmark across 12 multivariate datasets comparing 1D JEPAs against the 224-bit circuit-breaker panel plus the 130-stream baseline screen, Tables III(d) and V(a)). At $K = 25$ channels, online parameter counts are matched (272k for TS-JEPA, 312k for Soliton-Reynolds JEPA, 304k–307k for TCN observation forecasters, 649k for TimesNet, and 206k for TranAD; the 224-bit circuit-breaker panel adds zero trainable parameters). Runs executed on PyTorch 2.4 / CUDA 12.4 (RTX 5090, 0.60–0.88 ms/window, 188 MB memory); code is at https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [15], [39], we evaluate detectors across three tiers: (i) *primary accuracy and false-alarm outcomes* comprising realized held-out nominal FPR (FPR_{hold}), unadjusted Point-F1, and threshold-free PR-AUC (computed from the total active switch count $N_{\text{tot}}(\tau)$ for the 224-bit panel); (ii) *operating characteristics* evaluating event recall across autonomous circuit-breaker operating points (*224-Bit Strict* and *224-Bit Breaker*), deployable calibration quantiles (q^*, Fig. 4(b)), and post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(c)); and (iii) *secondary operational endpoints* reporting unweighted Dataset-Macro Point-F1, event recall (ER, fraction of contiguous anomaly segments with ≥ 1 alarm), mean detection delay $\Delta t = t_{\text{alm}} - t^*$ over detected events, and the constant-alarm reference $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Perspective Expansion Ladder and Circuit-Breaker Component Analysis

How do each of the seven fluid-dynamic perspectives and three calibration rules contribute to the 224-bit circuit-breaker panel on the identical trained backbone? Table II(a),(b) and Fig. 3(a) evaluate the step-by-step perspective expansion ladder and leave-one-out component ablations across all 19 non-GHL streams (57 runs):

1) *From 1D scalar collapse to 32-bit and 128-bit sub-horizon switches (Table II(a)):* Collapsing the 128 whitened coordinates into a single 1D score (whether modulated by R_{LT} at $\alpha_{\text{sol}} = 0.12$ or unmodulated at $\alpha_{\text{sol}} = 0$, which satisfy TOST equivalence within $\Delta = \pm 0.020$, $p_{\text{TOST}} = 2.7 \times 10^{-5}$) allows one or two drifting coordinates to inflate the scalar threshold, restricting causal Point-F1 to 0.052 and delay to 246.6 steps. Replacing 1D collapse with 32 independent switches on the terminal horizon ($m = 4$) nearly doubles causal Point-F1 to 0.101, and expanding across all four nested sub-horizons (*128-Bit Sub-Horizons*) lifts causal Point-F1 to 0.129, buffered Point-F1 from 0.137 to 0.192, and buffered PR-AUC from 0.370 to 0.402.
2) *Adding Reynolds mean-flow drift, turbulent fluctuation, and convective right-edge acceleration:* Appending the 32 Reynolds mean-flow drift dials in *160-Bit* cuts causal detection delay in half from 146.2 to 69.1 steps and

TABLE II
MULTI-PERSPECTIVE CIRCUIT-BREAKER EXPANSION LADDER, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY (n=19 STREAMS).

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.052	.137	.291	.370	.398	.377	246.6	402.7
32-Bit Full Horizon (m=4)	.101	.119	.266	.366	**.643**	.236	114.4	128.8
128-Bit Sub-Horizons (m=1..4)	.129	.192	.310	.402	.590	.525	146.2	514.5
160-Bit (+32 Reynolds Drift)	.132	.188	.311	.404	.585	.530	**69.1**	112.4
192-Bit (+32 Reynolds Fluc.)	.131	.203	**.312**	.408	.575	.567	179.0	217.3
224-Bit Breaker (+32 Accel./Relays)	**.138***	**.228**†	**.312**	**.411**	.607	**.631**	105.9	**71.8**

* Wilcoxon vs. 1D Scalar Collapse: 17/0 wins (p=1.5×10^{-5}); † buffered F1 p=0.026. Fixed-backbone 1D α_{sol}=0.12 vs. α_{sol}=0 (.055/.138 F1) satisfies TOST at Δ= ± 0.020 (p_{TOST}=2.7×10^{-5}).

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	**.138**	**.228**	.312	.411	**.607**	**.631**	105.9	**71.8**
224-Bit Strict (High Quorum K_{strict})	.116	.190	.312	.411	.534	.545	248.0	97.3
224-Bit + Soft Attention	.137	.226	.311	.410	**.607**	.628	106.8	73.4
w/o 32 Right-Edge Relays	.135	**.228**	**.313**	**.413**	.602	**.631**	123.7	72.5
w/o Prefix Routing (Unif. Smear)	**.138**	.223	.312	.407	**.607**	.584	105.9	88.6
w/o Hijack Guards (γ_j=1)	.137	.225	.312	.411	.603	.623	**101.6**	75.6

Leave-one-out component ablations evaluated on the identical 57 cached Soliton-Reynolds multi-horizon residual fields (n=19 streams × 3 seeds).

(c) Matched-Backbone Architectural Comparison (n=19 Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Soliton-Reynolds 224-Bit Breaker	2.30	0.179	0.017	0.132	0.138	0.228	**0.312**	**0.411**	**0.607**	0.774	**0.584**	**0.728**
Soliton-Reynolds 224-Bit Strict	2.30	0.179	0.011	0.098	0.116	0.190	**0.312**	**0.411**	0.534	0.774	**0.584**	**0.728**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Soliton-Reynolds (1D Scalar)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Soliton-Reynolds (α_{sol}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Hard-Target Copy	1.16	**0.167**	0.009	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

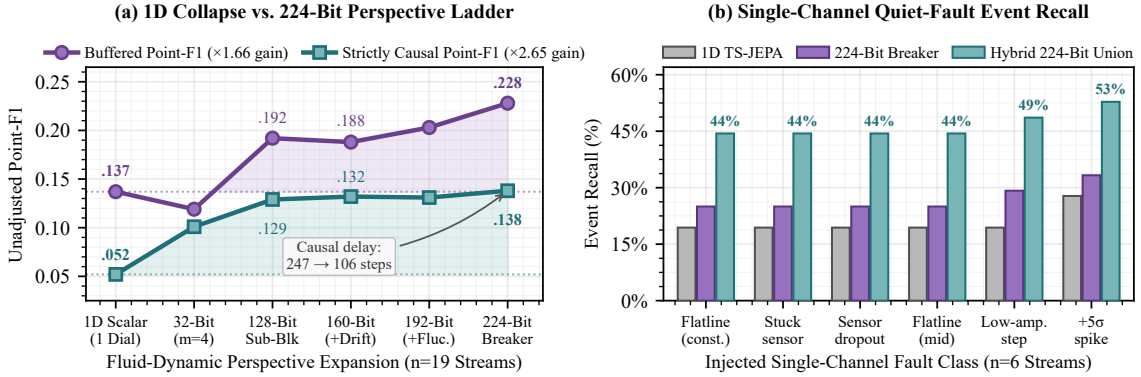


Fig. 3. Perspective expansion and quiet-fault recovery. **(a)** Causal and buffered Point-F1 across the 6-step fluid-dynamic expansion from 1D scalar collapse to the 224-bit circuit-breaker panel (n=19 streams). **(b)** Single-channel fault event recall comparing 1D TS-JEPA, standalone 224-Bit Breaker, and the Hybrid 224-Bit Union (n=6 streams).

lifts causal Point-F1 to 0.132. Adding the 32 Reynolds turbulent-fluctuation dials in *192-Bit* lifts buffered Point-F1 to 0.203 and buffered ER to 56.7%. Finally, adding the 32 convective right-edge acceleration dials and 32 onset relays in *224-Bit Breaker* reaches 0.138 causal Point-F1 (17 wins / 0 losses over 1D collapse, $p = 1.5\times10^{-5}$), 0.228 buffered Point-F1 ($p = 0.026$), 0.411 buffered PR-AUC, 60.7% causal and 63.1% buffered ER, and 71.8-step buffered delay.

3) *Leave-one-out component ablations (Table II(b)):* Removing the 32 fast right-edge onset relays slows causal detection delay from 105.9 to 123.7 steps and lowers causal Point-F1 from 0.138 to 0.135. Replacing physical prefix-horizon routing with uniform 64-step window smearing degrades buffered ER from 63.1% to 58.4%, buffered Point-F1 from 0.228 to 0.223, and buffered delay from 71.8 to 88.6 steps. Removing per-dial hijack guards ($\gamma_j = 1$) lowers causal and buffered F1 to 0.137 and 0.225, whereas adding a soft attention pooling gate provides no gain over discrete Boolean switches (0.137 causal and 0.226 buffered F1).

B. Ablation 2: Single-Channel Quiet Faults and the 224-Bit Hybrid Variance-Collapse Monitor

When one sensor out of K freezes at a nominal level while the remaining $K-1$ channels fluctuate normally, the frozen channel perturbs the pooled pre-LayerNorm vector by only $O(1/K)$, and LayerNorm rescales the embedding onto $\sqrt{D}\mathbb{S}^{D-1}$ using the remaining active channels. Pre-weighting input channels by context variance degrades macro Point-F1 across multivariate streams (0.118 → 0.077), so the deployed pipeline retains uniform input encoding. Across six single-channel fault classes injected into six multivariate streams (Fig. 3(b)), 1D TS-JEPA detects only 19.4% of quiet-fault events (flatlines, stuck sensors, and dropouts) with $< 3\%$ point recall. Because the 224-bit circuit-breaker panel monitors all 32 latent coordinates independently without 1D score collapse, standalone *Soliton-Reynolds 224-Bit Breaker* lifts

quiet-fault event recall to 25.0%–29.2% (33.3% on $+5\sigma$ spikes) and increases point recall more than seven-fold to 14.3%–25.6%. Pairing the 224-bit panel in a logical union with a parameter-free observation-space rolling variance-collapse monitor (tracking how far each channel's 32-step trailing standard deviation drops below its nominal median) lifts event recall to 44.4%–52.8% and point recall to 24.3%–35.6% across all six single-channel fault classes.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

To isolate the effect of the 4-shell rFFT front-end under the 224-bit circuit-breaker panel, we train Soliton-Reynolds JEPA with and without dyadic frequency-shell splitting across five benchmark streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, and `GECCO water-quality`, three seeds). On high-dimensional spacecraft telemetry with multi-frequency actuator oscillations, the dyadic front-end raises Point-F1 and PR-AUC: on `MSL M-1` ($K = 55$), buffered Point-F1 jumps from 0.085 to 0.410 and PR-AUC from 0.505 to 0.757, and on `SMAP P-3` ($K = 25$), causal Point-F1 rises from 0.212 to 0.321 and PR-AUC from 0.367 to 0.415. It also cuts mean detection delay across the five streams from 28.6 to 26.2 steps causally and from 23.5 to 17.9 steps buffered while lowering validation residual kurtosis from 1.41 to 1.07. Conversely, on lower-dimensional kinematic and water-quality streams ($K = 9$), a plain time-domain convolutional stem achieves higher Point-F1 (0.217 vs. 0.136 on `Daphnet` and 0.321 vs. 0.158 on `GECCO`), yielding a 5-stream macro causal Point-F1 of 0.193 for the standard stem versus 0.156 for the dyadic stem. Frequency-shell decomposition is thus most beneficial on high-channel telemetry with distinct spectral bands.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 15 model arms across the 19 non-GHL streams under identical splits, training budgets, and matched parameter counts in Table II(c) isolates five architectural factors:

1) *Forecasting vs. reconstructing raw observations (TCN Obs. Forecaster vs. TCN Obs. Autoencoder):* Pairing the same causal dilated TCN encoder with a multilayer perceptron decoder to either forecast the 64-step target window or reconstruct the 320-step window in raw sensor space leaves tail kurtosis ($\gamma_2 \approx 6.2$), calibration drift ($D_{\mathrm{KS}} \approx 0.45$), and causal FPR ($\approx 25\%$) nearly identical.
2) *LayerNorm bottleneck in observation space (TCN Obs. Forecaster (LN)):* Adding a $D = 32$ LayerNorm bottleneck and target-window LayerNorm to the observation forecaster compresses tail kurtosis six-fold ($\gamma_2 = 0.91$) and halves causal FPR to 11.7%, but targeting raw sensor waveforms leaves calibration drift high ($D_{\mathrm{KS}} = 0.376$) and drops causal Point-F1 to 0.084.
3) *Learned latent target projection, multi-horizon whitening, and 224-bit circuit breakers (TS-JEPA and Soliton-Reynolds 224-Bit):* Moving the prediction target into the learned $D = 32$ latent space cuts calibration drift nearly in half ($D_{\mathrm{KS}} = 0.214$ in TS-JEPA and 0.179 in Soliton-Reynolds JEPA, paired Wilcoxon $p \leq 0.001$), while the 224-bit circuit-breaker panel lifts buffered PR-AUC to 0.411, buffered Point-F1 to 0.228, and causal event recall to 0.607 at 1.7% held-out FPR.
4) *Nonlinear TCN encoding vs. linear PCA and tree isolation (PCA Linear Forecaster (LN) and Isolation Forest):* Replacing the nonlinear TCN with 32 principal components and ridge regression bounds causal FPR to 2.1% through LayerNorm, but causal Point-F1 collapses to 0.017 (and Isolation Forest reaches only 0.149 causal event recall), confirming the need for learned nonlinear temporal features.
5) *Future prediction vs. context autoencoding, EMA stability, and ℓ_2 scoring (JEPA Context Autoencoder, JEPA Hard-Target Copy, and TS-JEPA (ℓ_2 Scoring)):* Context autoencoding in latent space keeps causal FPR low (1.7%) but drops causal Point-F1 from 0.161 to 0.112 and PR-AUC from 0.278 to 0.235. Direct weight copying ($\zeta = 0$) or Euclidean ℓ_2 scoring still holds causal FPR below 2% (0.9% and 1.8%), confirming that false-alarm reduction stems from the normalized latent representation.

VII. Results and Discussion

A. Primary 33-Stream and Archive-Wide 141-Stream Benchmark Results

Table III evaluates all 33 primary continuous telemetry streams under the two-stage chronological held-out protocol: Panel (a) reports retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns including anomaly event counts (E) and held-out nominal FPRs. Archive-wide 141-stream results appear in Table V; on a 130-stream baseline screen, TS-JEPA achieves 0.186 stream and 0.160 macro Point-F1 versus 0.152 for TranAD, 0.151 for TimesNet, 0.097 for DCdetector, 0.088 for NCAD, and 0.082 for Anomaly Transformer. Across both attribution regimes, pairing multi-horizon Soliton-Reynolds JEPA with the 224-bit fluid-dynamic circuit-breaker panel achieves the highest buffered Point-F1 (0.228), buffered PR-AUC (0.411), causal PR-AUC (0.312), causal event recall (0.607), and Dataset-Macro Point-F1 (0.122), while keeping held-out false alarms an order of magnitude below observation-space reconstructors (1.1%–1.7% causal and 9.8%–13.2% buffered vs. 22.2%–30.7% for TimesNet and TranAD).

Eliminating Observation-Space False-Alarm Flooding: On the 19 non-GHL primary streams (Table III(a),(b)), predicting spherical latent representations cuts held-out nominal FPR three-fold under buffered attribution to 7.8%–9.8% versus 30.0%–30.7% for TimesNet and TranAD ($p \leq 0.045$), and by an order of magnitude under strictly causal streaming to 0.6%–1.7% versus 22.2%–24.3% for reconstructors ($p = 0.010$). This separation holds across all $N = 7$ dataset families ($p = 0.016$) and under zero-tuning LODO evaluation, where JEPAs maintain 0.3%–2.3% causal FPR versus 12.1%–18.4% for reconstructors. In Table III(c), Soliton-Reynolds 224-Bit achieves 0.7% causal and 4.7% buffered FPR on Daphnet with 0.223 buffered F1 versus 0.145 for TS-JEPA, leads all methods

TABLE III
HELD-OUT MULTI-STREAM TELEMETRY BENCHMARK ACROSS ATTRIBUTION REGIMES AND DATASETS (BEST IN **BOLD**, SECOND UNDERLINED).

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-perspective fluid-dynamic circuit-breaker architectures												
Soliton-Reynolds 224-Bit Breaker	0.132	**0.228**	**0.411**	**0.631**	71.8	0.017	0.138	**0.607**	105.9	**0.002**	0.000	**0.122**
Soliton-Reynolds 224-Bit Strict	0.098	0.190	**0.411**	0.545	97.3	0.011	0.116	0.534	248.0	**0.002**	0.000	0.098
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Soliton-Reynolds (1D scalar)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Soliton-Reynolds (α_{sol}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	F1_{all}	**224-Bit**	**TS-JEPA**	**Sol. 1D**	**TimesNet**	**TranAD**	**224-Bit**	**TS-JEPA**	**Sol. 1D**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.223	0.145	0.143	**0.229**	0.144	**.007/.047**	.030/.091	.008/.072	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.001/.002**	.004/.004	**.001/.002**	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	**0.063**	0.000	0.012	0.000	0.000	.010/.038	.004/.017	.003/.098	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.199**	0.161	0.081	0.158	0.151	.014/.131	.054/**.095**	**.004**/.098	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.354	**0.379**	0.267	0.289	0.167	.021/.214	.014/.135	**.006/.093**	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	**0.001**	**0.001**	0.000	.002/.007	.008/.014	.002/.007	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.017	**0.119**	0.008	0.000	0.000	**.000/.000**	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**

Note: In (c), n = streams, K = channels, E = events, π = anomaly rate, $\text{F1}_{\text{all}} = 2\pi/(1+\pi)$. 95% stream-bootstrap CIs (n=19): 224-Bit Breaker buffered FPR [.068, .198], F1 [.142, .321] and causal FPR [.008, .027], F1 [.068, .224] (vs. Sol.-Rey. 1D causal F1 [.013, .097] and TimesNet causal FPR [.123, .375]).

on MSL at 0.199 buffered F1 versus 0.161 for TS-JEPA and 0.158 for TimesNet, reaches 0.354 buffered and 0.351 causal F1 on SMAP at a 38.2-step delay, and reaches 0.063 buffered F1 on Genesis where reconstructors score 0.000.

Breaking the 1D Scalar Score-Collapse Bottleneck: Under legacy 1D scalar scoring, Soliton-Reynolds JEPA trades lower Point-F1 (0.050 causal, 0.148 buffered) for ultra-conservative firing because one or two out-of-sample drifting coordinates (*hijacker dials*) inflate the scalar threshold. Replacing 1D scalar collapse with the 224-bit multi-perspective circuit-breaker panel resolves this trade-off on the exact same trained networks (Tables II(a), III(a),(b), and V). On the 19 primary streams, *Soliton-Reynolds 224-Bit Strict* and *224-Bit Breaker* lift buffered PR-AUC from 0.331–0.370 to 0.411, raise buffered Point-F1 to 0.190–0.228 ($p = 0.026$), more than double causal Point-F1 from 0.052 to 0.116–0.138 (17 wins / 0 losses, $p = 1.5\times10^{-5}$), lift causal event recall to 53.4%–60.7% and buffered recall to 63.1%, and cut detection delay from 246.6 to 105.9 steps causally and from 402.7 to 71.8 steps buffered. On GHL ($\pi = 0.013$, Table III(c)), all JEPAs and the linear autoregressive (AR) forecaster fire near-silently at 0.1%–0.4% FPR with 0.000 F1, confirming that low FPR without recall does not indicate detection.

Operational Applicability Regimes: Performance across fault types and sampling rates defines four operational regimes: (i) *multi-channel spacecraft, server, and wearable arrays* with $K \geq 9$ across SMAP, MSL, Daphnet, SMD, Exathlon, and OPPORTUNITY, where the 224-bit panel maximizes Point-F1 and event recall at low false-alarm rates; (ii) *latency-constrained infrastructure* where a 63-step window lag is prohibitive, such as hourly Metro where 63 steps = 63 h, requiring short horizons $S \in \{1, 4\}$ (Section VII-D); (iii) *quiet single-channel sensor freezes* absorbed by LayerNorm, where pairing the 224-bit panel with the rolling variance-collapse monitor recovers 44.4%–52.8% of events (Section VI-B); and (iv) *continuous non-stationary process ramps* on GHL and Metro (0.000–0.022 F1), which defeat static calibration rules.

B. Synthesis of Pipeline Stages: Barycenter Tapping, Multi-Horizon Whitening, and 224-Bit Perspectives

Tables II(a)–(c) and III(a),(b) trace how the four pipeline stages from Section IV drive detection performance:

1) *Stage 1 (Multi-depth TCN tapping and synchronized barycenter centering):* Subtracting the online context centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}}$ removes common-mode level shifts before spherical projection, lowering calibration-to-holdout drift D_{KS} from 0.214 in TS-JEPA to 0.179 in Soliton-Reynolds JEPA (Table II(c)).
2) *Stage 2 (Nested multi-horizon rollout and whitening defect regularization):* Rolling out predictions across $M = 4$ prefixes ($S_m \in \{16, 32, 48, 64\}$) under the Frobenius whitening defect objective (2) raises $\text{RankMe}(z_{\text{tgt}})$ from 11.6 to 19.8 and lifts causal event recall at $f^* = 0.01$ from 0.395 to 0.584.
3) *Stages 3–4 (Calibration-stabilized whitening, 224-dial Reynolds expansion, and onset relays):* Post-hoc 1D Lieb–Thirring modulation ($\alpha_{\text{sol}} = 0.12$) leaves 1D Point-F1 unchanged under TOST ($p_{\text{TOST}} = 2.7\times10^{-5}$, Table II(a)); by contrast, expanding the 32 whitened coordinates into $7 \times 32 = 224$ hijack-guarded switches and 32 right-edge onset relays eliminates cross-coordinate dilution, doubling causal Point-F1 and halving detection delay.

C. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table III shows why the 224-bit circuit-breaker panel is essential for real-time streaming. Without 64-step window averaging, causal 128-dimensional whitened residuals exhibit a heavier right tail ($\gamma_2 = 2.30$) than buffered scores ($\gamma_2 = 1.25$). Under legacy 1D scalar scoring, hijacker dials push the scalar threshold onto that tail so that 1D Soliton-Reynolds JEPA alerts on only

TABLE IV
CONTEXTUAL VS. POINT ANOMALY DETECTION (EVENT RECALL (ER)).

	NASA Contextual (n=5 seqs)					NASA Point (n=8)	
(a) Model	**Buf. PR**	**Auto. ER**	**FPR**	f^*=.01	f^*=.05	f^*=.01	f^*=.05
Sol.-Rey. 224-Bit	**0.658**	**0.750**	0.018	**0.820**	**0.920**	0.683	0.944
Soliton-Rey. (1D)	0.652	**0.750**	**0.006**	**0.820**	**0.920**	0.683	0.867
Sol.-Rey. (α_{sol}=0)	0.641	0.625	**0.006**	**0.820**	**0.920**	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

	19-Stream Contextual ($\lvert z\rvert \leq 5$)			19-Stream Point ($\lvert z\rvert > 5$)		
(b) Model	q^*=0.05	f^*=0.01	f^*=0.05	q^*=0.05	f^*=0.01	f^*=0.05
Sol.-Rey. 224-Bit	**0.654**	**0.552**	**0.618**	**0.796**	0.688	**0.792**
Soliton-Rey. (1D)	0.628	0.498	0.590	0.782	0.683	0.781
Sol.-Rey. (α_{sol}=0)	0.640	0.540	0.603	0.785	0.671	0.785
TS-JEPA	0.614	0.520	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Panel (a) stratifies the 10 primary SMAP/MSL streams by NASA's ground-truth taxonomy [4] (contextual: SMAP E-1, A-2, A-7, MSL M-1, and C-1 [2100, 2210]; point: SMAP A-1, P-3, D-3, MSL D-14, D-15, and C-1 [550, 750]). Panel (b) stratifies all n=19 non-GHL streams into in-range contextual ($\max|z| \leq 5\sigma$) vs. out-of-range point ($\max|z| > 5\sigma$) causal events.

$\hat{p}_{\text{alert}} = 2.4\%$ of test steps (4.0% true-positive rate (TPR) at 0.6% FPR), flagging only isolated peak steps inside each anomaly (0.050–0.052 causal F1). By monitoring each fluid-dynamic dial with its own hijack-guarded switch $b_j(t)$ and firing immediately when a terminal sub-block and right-edge convective acceleration relay $r_c(t)$ coincide in (7), *224-Bit Breaker* stays active across the true fault window, reaching 0.138 causal F1, 60.7% event recall, and 105.9-step delay at 1.67% FPR. Against an analytical rate-matched i.i.d. Bernoulli random alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, random firing at 224-Bit Breaker's held-out nominal rate catches only 28.1% of events versus 60.7% actual (27.2% vs. 47.8% for 1D Soliton-Reynolds and 28.3% vs. 48.7% for TS-JEPA), whereas for TimesNet (24.3% FPR) random firing alone reaches 60.0%, exceeding TimesNet's actual 58.8%.

D. Horizon Length, Structural Emission Lag, and Fast-Tripwire Regimes

Target horizon $S = 64$ imposes a structural emission lag of $S - 1 = 63$ steps before a causal trailing window decision is emitted at $\tau = t + S - 1$ (Section IV-D). To evaluate detection quality versus latency against baselines at matched emission lag $S - 1 \in \{0, 3, 15, 31, 63\}$, Fig. 4(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams at three seeds. At $S = 1$ (zero lag) and $S = 4$ (3-step lag), latent prediction keeps causal FPR at 0.4%–0.6% versus 2.5%–3.1% for TimesNet, with higher causal PR-AUC (0.178–0.189 vs. 0.150–0.162) and short delays (Δt = 57.8–85.2 steps). As S increases from 16 to 64, TimesNet causal FPR climbs from 6.1% to 21.3%, whereas latent predictors remain bounded at $\leq$ 2.0%. Thus, $S = 64$ suits high-rate telemetry ($\geq$ 1 Hz, such as 64 Hz Daphnet where 63 steps = 0.98 s) or subsystem trend monitoring on 0.1 Hz SMAP and MSL (10.5 min), while low-rate streams ($\leq 1\,\text{min}^{-1}$) and zero-lag tripwires require $S \in \{1, 4\}$.

E. Operating Budgets and Contextual vs. Point Anomalies

Because reconstruction baselines calibrate to 22%–31% realized FPR versus 0.6%–2.4% for latent predictors, we evaluate three operating-point comparisons:

TABLE V
ARCHIVE-WIDE 141-STREAM CIRCUIT-BREAKER BENCHMARK (12 DATASETS, 3 SEEDS) AND PER-FAMILY BREAKDOWN.

	Pt-F1 ↑		PR-AUC ↑		ER ↑		Delay Δt ↓	
(a) All n=141 Streams	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**	**Caus.**	**Buf.**
TS-JEPA (1D)	.067	.142	.165	.213	.480	.459	115.4	117.9
Speculative-JEPA (1D)	.071	.155	.183	.256	.497	.565	123.0	64.8
Spec.-Alien-JEPA (1D)	.077	.153	.184	.253	.507	.567	125.4	65.1
Spec.-224Bit-Strict	.125*	.177‡	**.197**	**.261**	.648	.660	78.0	54.0
Spec.-224Bit-Breaker	**.139**†	**.180**‡	**.197**	**.261**	**.727**	**.685**	**55.3**	**46.6**

		Causal Pt-F1 ↑			Buffered Pt-F1 ↑			Causal ER ↑		
(b) Family	n	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**	**224B**	**1D-Al.**	**TS**
SMAP	47	**.202**	.130	.080	**.219**	.213	.218	**.857**	.709	.595
Daphnet	22	**.097**	.038	.064	**.155**	.123	.098	**.863**	.550	.595
MSL	18	**.131**	.044	.064	**.278**	.183	.170	**.787**	.574	.586
SMD	18	**.090**	.028	.042	**.125**	.065	.072	**.234**	.106	.088
Exathlon	17	**.163**	.133	.092	.181	**.186**	.134	**.821**	.632	.472
OPPORT.	13	**.075**	.017	.045	**.066**	.062	.036	**.821**	.231	.487

* Wilcoxon vs. 1D Spec.-Alien $98/29$ wins ($p = 1.8\times10^{-11}$); † $114/18$ wins ($p = 1.2\times10^{-16}$) and $97/36$ vs. TS-JEPA ($p = 8.3\times10^{-11}$); ‡ buffered F1 $p \leq 2.6\times10^{-4}$. In (b), 224B = Spec.-224Bit-Breaker, 1D-Al. = 1D Spec.-Alien, TS = TS-JEPA (n=135 multi-stream; 6 single-stream datasets in text).

1) *Autonomous circuit-breaker points and nominal calibration quantiles (q^*, Fig. 4(b), Table II(c)):* Without test labels, Soliton-Reynolds 224-Bit Strict (53.4% ER at 1.1% FPR) and 224-Bit Breaker (60.7% ER at 1.7% FPR) dominate the low-FPR frontier; at calibration quantile $q^* = 0.050$, Soliton-Reynolds 224-Bit reaches 77.4% causal ER versus 61.8% for TimesNet and 60.2% for TranAD.
2) *Post-hoc matched realized test-FPR budgets (f^*, Fig. 4(b), Table II(c)):* Equalizing realized test nominal FPR across models (vertical dotted guides in Fig. 4(b)) proves that low false-alarm rates are not an artifact of low alert frequency. At a 1% test FPR budget ($f^* = 0.01$), Soliton-Reynolds 224-Bit detects 58.4% of anomaly events (56.2%–57.8% for 1D Soliton-Reynolds and 39.5% for TS-JEPA), nearly tripling TimesNet (21.7%) and quadrupling TranAD (14.9%), with a similar gap at $f^* = 0.05$ (72.8% vs. 34.3%–40.7%).
3) *Contextual vs. point anomaly stratification (Table IV):* Out-of-range point anomalies ($\pm5\sigma$ spikes) are flagged even by a marginal z-score (`ZScoreMax`), whereas contextual anomalies stay within normal bounds ($\max|z| \leq 5\sigma$) and violate multi-step trajectory structure. In Table IV(a), stratifying the 10 primary SMAP/MSL streams by NASA's taxonomy [4] shows that observation-space detectors collapse on contextual anomalies (0.161–0.317 buffered PR-AUC and 30%–56% recall at $f^* \in \{0.01, 0.05\}$). Soliton-Reynolds 224-Bit leads every contextual metric with 0.658 buffered PR-AUC, 75.0% autonomous recall, and 82.0%–92.0% recall at $f^* \in \{0.01, 0.05\}$, and reaches 94.4% recall at $f^* = 0.05$ on NASA point anomalies. Across all 19 non-GHL streams in Table IV(b), Soliton-Reynolds 224-Bit doubles reconstructor recall on contextual faults (61.8% vs. 30.0% at $f^* = 0.05$) and leads on point faults (79.2% vs. 56.3%–61.1%).

F. Archive-Wide 141-Stream Analysis, Hijack Isolation, and Drift Decomposition

Table V evaluates the 224-bit circuit-breaker panel across all 141 qualifying multivariate streams in 12 datasets at three seeds (423 runs per model):

1) *Archive-wide gains over 1D single- and multi-horizon JEPAs (Table V(a)):* Moving from single-horizon TS-JEPA

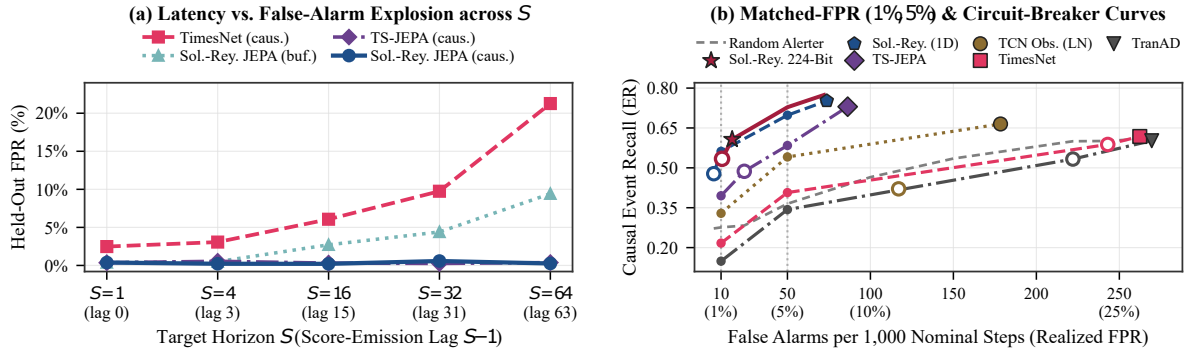


Fig. 4. Horizon length and operating trade-offs. **(a)** Held-out FPR across target horizons $S \in \{1, 4, 16, 32, 64\}$ at matched emission lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps (n=5 streams). **(b)** Causal event recall vs. false alarms across n=19 streams at matched test-FPR budgets $f^* \in \{1\%, 5\%\}$ (vertical dotted guides), autonomous 224-Bit Strict (open circle) and 224-Bit Breaker (filled star) operating points, and calibration quantiles $q^* \leq 0.05$.

to 3-horizon Speculative-JEPA and Speculative-Alien-JEPA under 1D scoring first lifts PR-AUC from 0.165 to 0.184 causally and from 0.213 to 0.253–0.256 buffered ($p \leq 0.0042$). Pairing the multi-horizon predictor with the 224-bit circuit-breaker panel in *Speculative-224Bit-Strict* and *Speculative-224Bit-Breaker* then lifts causal PR-AUC to 0.197 ($p = 0.014$ vs. TS-JEPA), nearly doubles causal Point-F1 from 0.067–0.077 to 0.125–0.139 (97/36 wins over TS-JEPA, $p = 8.3\times10^{-11}$; 114/18 wins over 1D Speculative-Alien-JEPA, $p = 1.2\times10^{-16}$), raises causal event recall from 48.0%–50.7% to 64.8%–72.7%, and halves causal detection delay from 115.4–125.4 to 55.3–78.0 steps. Under buffered attribution, prefix-routed voting lifts PR-AUC to 0.261 ($p = 0.0031$), raises Point-F1 to 0.177–0.180 ($p \leq 2.6\times10^{-4}$), lifts event recall to 66.0%–68.5%, and lowers held-out FPR below 1D Speculative-Alien-JEPA (16.5% vs. 18.0%) at 46.6–54.0-step delay.

2) *Consistency across all dataset families (Table V(b)):* Across every multi-stream family, *Speculative-224Bit-Breaker* leads causal Point-F1 and event recall. On SMAP ($n = 47$), causal Point-F1 rises from 0.080 and 0.130 to 0.202 at 85.7% ER; on Daphnet ($n = 22$), it reaches 0.097 causal and 0.155 buffered F1 at 86.3% ER; on MSL ($n = 18$), it reaches 0.278 buffered and 0.131 causal F1 versus 0.170–0.183 buffered for 1D JEPAs; on SMD ($n = 18$), causal F1 doubles from 0.028–0.042 to 0.090 (0.125 buffered); and on Exathlon ($n = 17$) and OPPORTUNITY ($n = 13$), causal ER climbs to 82.1% versus 47.2%–48.7% for TS-JEPA. On the six single-stream datasets, GECCO reaches 0.149 causal and 0.207 buffered F1 (7.2% ER), SWAN reaches 0.578 buffered PR-AUC (0.240 F1 vs. 0.106 for 1D Spec.-Alien), and Genesis, PSM, CreditCard, and Metro reach 66.7%, 7.7%, 25.6%, and 4.0% buffered ER.

3) *Three-stage permutation drift decomposition:* Refitting thresholds across the 19 primary streams under chronological (FPR_{chr}), $L = 64$ block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq 0.35\%$) splits confirms that inter-block regime drift ($\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$) drives 71.3% of excess causal FPR in TS-JEPA (0.0229 vs. 0.0073) and 78.3%–81.8% in reconstructors (0.210–0.230 vs. 0.043–0.046), whereas barycenter centering in Soliton-Reynolds JEPA cuts the inter-block drift share four-fold to 18.7% (0.0056 vs. 0.0047) while per-dial hijack guards γ_j quarantine individual drifting coordinates.

VIII. Conclusion and Future Work

We investigated how Joint-Embedding Predictive Architectures and fluid-dynamic multi-perspective circuit-breaker panels resolve both observation-space false-alarm flooding and 1D scalar score collapse across 33 primary and 141 archive-wide continuous telemetry streams:

1) *224-Bit Fluid-Dynamic Circuit Breakers Eliminate 1D Score Collapse:* Expanding $D = 32$ whitened coordinates into $7 \times 32 = 224$ fluid-dynamic dials across sub-horizon positions, Reynolds mean-flow transport, turbulent fluctuation, and convective right-edge acceleration with hijack guards, prefix-routed attribution, and 32 right-edge onset relays prevents drifting coordinates from inflating scalar thresholds. On n=19 primary streams, this 6-step expansion lifts buffered Point-F1 to 0.228 at 0.411 PR-AUC and 71.8-step delay, doubles causal Point-F1 from 0.052 to 0.138 (17/0 wins, $p = 1.5\times10^{-5}$), and halves causal delay from 246.6 to 105.9 steps; across all 141 streams (423 runs), it doubles causal Point-F1 from 0.067–0.077 to 0.139 (114/18 wins, $p = 1.2\times10^{-16}$) at 72.7% recall and 55.3-step delay.

2) *Factorial Mechanism Isolation and Component Verification:* Matched-encoder ablations isolate LayerNorm kurtosis compression (γ_2 : 6.10 $\rightarrow$ 0.91), learned latent projections (D_{KS} : 0.376 $\rightarrow$ 0.214), and multi-horizon barycenter whitening in Soliton-Reynolds JEPA (RankMe : 11.6 $\rightarrow$ 19.8, $D_{\text{KS}} = 0.179$, inter-block drift share 71.3% $\rightarrow$ 18.7%). Leave-one-out ablations confirm the 32 right-edge relays (-17.8-step causal delay), prefix routing ($+4.7\%$ buffered recall), and hijack guards, while post-hoc 1D Lieb–Thirring modulation satisfies TOST equivalence ($p_{\text{TOST}} < 10^{-4}$) and short horizons ($S \in \{1, 4\}$) eliminate the 63-step emission lag.

3) *Contextual-Anomaly Dominance and Hybrid Quiet-Fault Coverage:* Soliton-Reynolds 224-Bit nearly triples reconstructor recall at a 1% test-FPR budget (58.4% vs. 21.7% for TimesNet) and leads on NASA contextual anomalies

(0.658 buffered PR-AUC; 82.0%–92.0% matched-FPR recall), while standalone 224-Bit Breaker lifts single-channel freeze point recall 7-fold (14.3%–25.6%) and its union with a sensor-variance monitor reaches 44.4%–52.8% event recall, motivating online conformal circuit-breaker adaptation for non-stationary ramps.

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